Dev Shah

Research interests: agentic data science, evaluation of LLM agents, causal discovery, natural language processing.

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EDUCATION

University of Maryland, College Park — College Park, MD
M.S. in Data Science | GPA: 4.0 / 4.0 — *Aug 2025 – May 2027*
Coursework: Machine Learning, Probability & Statistics, Algorithms for Data Science, Agentic AI, Big Data Systems

Indian Institute of Information Technology, Lucknow — Lucknow, India
B.Tech in Computer Science & Artificial Intelligence | GPA: 8.78 / 10 — *Nov 2021 – May 2025*
Coursework: Data Structures & Algorithms, Artificial Intelligence, NLP, Generative AI, DBMS, Operating Systems

RESEARCH EXPERIENCE

University of Maryland – Agentic Data Science — College Park, MD
Student Researcher · Advisor: Dr. Giacinto Paolo Saggese — *Jul 2026 – Present*

- Building a framework that evaluates LLM agents against known ground truth: a synthetic data generator produces the answer key, an agent driver runs the method, and a scoring pipeline grades it. The harness is method-agnostic, so a new method plugs into one stage.
- First benchmark is causal discovery: recovered graphs are scored against the true DAG by Structural Hamming Distance (SHD) and edge precision/recall. A scripted baseline beats PC and GES at 5 nodes and stays within 1.4 SHD of the better of them up to 10 nodes.
- Rebuilt the agent loop after finding the prototype computed answers in Python while the model only formatted them, so scores measure the model, not the scaffolding. Kept DAG-vs-CPDAG scoring a parameter, since Gaussian data identifies only the equivalence class.
- Authored a survey of automated EDA (~60 tools and papers in five approach families) and the benchmark scoping document.

IIIT Lucknow – AI Research Lab — Lucknow, India
AI Research Intern · Advisor: Dr. Deepak Kumar Singh — *Jan 2024 – May 2024*

- Benchmarked YOLOv3–v8 and Faster R-CNN on a Raspberry Pi, profiling each model's accuracy/latency trade-off on the device itself.
- Found that the model ranking changes once memory and compute are constrained, so chose the deployable model on measured on-device behavior rather than offline score; improved inference speed 12%.

IIIT Allahabad – Research Lab — Remote
Research Intern · Advisor: Dr. Naveen Saini — *Aug 2023 – Nov 2023*

- Built a knowledge graph recommender on 10K+ multilingual ratings (18 languages); PCA and clustering improved precision 9%.
- Shipped a Flask REST API serving the recommender's predictions, so other applications could query it directly.

TEACHING

University of Maryland — College Park, MD
Teaching Assistant · MSML610: Advanced Machine Learning — *Aug 2026 – Dec 2026*

SELECTED PROJECTS

Phishing Email Detection — *BERT, Hidden Markov Models, scikit-learn*

- Found test-set leakage inflating the reported accuracy; rebuilt the pipeline leak-free and re-measured all 29,767 emails at 98.89%, 95% CI [98.62, 99.14], vs a 53.1% baseline.

Medical Text Summarization — *PyTorch, T5, Hugging Face Transformers*

- Fine-tuned T5 on 6,000 PubMed pairs, beating a lead-3 baseline by 1.84 ROUGE-L, 95% CI [1.06, 2.57] by paired bootstrap.

Multi-Source Data Reconciliation — *Python, SQL, DuckDB*

- Reconciled 144,422 records against a 429,637-record feed and showed the 97.5% match rate measured shared lineage, not agreement; traced 61.8% of real discrepancies to 2.4% of records.

INDUSTRY EXPERIENCE

Georgia Institute of Technology – Office of Development — Jun 2026 – Jul 2026
AI Strategy & Operations Intern: mapped workflows across 11 teams and scoped three AI agents for leadership

Nativebyte — Dec 2024 – May 2025
Software Developer Intern: built XGBoost and regression recommendation models end to end

SKILLS & ACHIEVEMENTS

Programming: Python, SQL, C++, R | **ML & NLP:** PyTorch, Hugging Face Transformers, scikit-learn, causal-learn, pgmpy

Statistics: hypothesis testing, bootstrap confidence intervals, experiment design | **Data:** PySpark, DuckDB, Docker, Git, AWS

Competitive Programming: CodeChef 3-star, best contest global rank 39 | LeetCode top 5%, best contest rank 931